

FEATURE SCALING

DEFINITION

Feature scaling is a preprocessing technique used to bring numerical features to a similar scale so that features with larger numerical values do not dominate the model.

WHY DO WE NEED FEATURE SCALING?

Consider example:

- Age → 20-60
- Salary → 20,000-2,00,000

Without scaling, Salary has much larger numerical values than Age. In algorithms that depend on distance or optimization, this can cause Salary to have a

disproportionately large influence.

Feature scaling helps put features on a comparable scale.

[Scale Contrast]

Age: [20 - 60] (small)

Salary: [20k - 2L] (huge)

↓ Distance bias

Salary dominates model!

↓ After Scaling

Both on comparable scale.

WHY IS IT IMPORTANT?

- Prevents large-scale features from dominating.
- Helps Gradient Descent converge faster.
- Important for distance-based algorithms.
- Makes optimization more efficient.

COMMON FEATURE SCALING TECHNIQUES

1. STANDARDIZATION

$$z = (x - \mu) / \sigma$$

After standardization:

- Mean ≈ 0
- Standard deviation ≈ 1

2. MIN-MAX SCALING

$$x' = (x - x_{\min}) / (x_{\max} - x_{\min})$$

Usually transforms values into the range 0 to 1.

3. ROBUST SCALING

- Uses median and IQR instead of mean and standard deviation.
- Useful when the dataset contains significant outliers

ALGORITHMS WHERE SCALING IS IMPORTANT

- KNN
- K-Means
- SVM
- Logistic Regression
- Linear Regression using Gradient Descent
- Neural Networks
- PCA

[Algorithm Groups]

Distance / Gradient based:

- Need Scaling (KNN, SVM, NN)

Tree-based:

- No Scaling needed!

ALGORITHMS WHERE SCALING USUALLY ISN'T REQUIRED

- *Decision Tree*
- *Random Forest*
- *XGBoost*
- *Other tree-based models*

Reason: *Tree-based algorithms make decisions using feature thresholds, rather than calculating distances or relying on gradient-based optimization.*

INTERVIEW ONE-LINER

Feature scaling is a preprocessing technique that brings numerical features to a comparable scale, preventing large-magnitude features from dominating and improving the performance or convergence of scale-sensitive machine learning algorithms.